INTEGRATION BEE 2025

Final

1. $$\int_0^{\infty} \sum_{n=1}^{+\infty} \left(2024e^{-2024nx} - 2025e^{-2025nx}\right) dx = \log\left(\frac{2025}{2024}\right)$$

2. $$\int_0^{+\infty} e^{-x^2} \cos(2025\ x)dx = \frac{\sqrt{\pi}}{2} e^{-\frac{2025^2}{4}}$$

3. $$\int_0^{\infty} \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15}$$

4. $$\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \frac{22}{7} - \pi$$

5. $$\int_0^{\infty} \sum_{k=\lfloor x \rfloor}^{\infty} \cos(k\pi) \frac{2025^{-k}}{(k+1)!} \pi^{2k+1} dx = \pi e^{-\frac{\pi^2}{2025}}$$